Juan Marulanda

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Education

Carleton University — Ottawa, ON

Bachelor of Computer Science, AI and Machine Learning Concentration — *2023 – 2027*

- **Relevant Coursework:** Data Analytics, Algorithms and Data Structures, Operating Systems, Compiler Construction, ML

Experience

Tatlock Systems — Ottawa, ON

Software Engineer Intern — *May 2026 – Aug 2026*

- Design and implement a memory-safe C dialect for microcontroller targets that eliminates pointer arithmetic, **eradicating** an entire class of buffer-overflow and dangling-pointer bugs at compile time.
- Developed an LSP server and VS Code extension for the language and for dialects of the **k language** providing syntax highlighting, hover documentation, and inline decoration.
- Architect a transpiler from the dialect to C, enabling deployment on existing embedded C compiler toolchains for 11 ESP32 chip variants.

Bedarra Corporation — Ottawa, ON

Software Engineer Intern — *May 2024 – Aug 2025*

- Engineered from scratch a **local-first cross-platform app** (iOS, Android, Web, macOS, Windows) in Flutter with a distributed **CRDT-based sync engine** in Rust, implementing store-and-forward propagation so distributed devices sync opportunistically on shared networks without central coordination.
- Implemented an offline-first **search engine** that synchronized local SQLite records to an on-device instance, enabling sub-millisecond responses for 800k+ records.
- Sped up email downloads from IMAP servers by 5x by compiling libraries to a portable runtime that enabled concurrent fetching, replacing a previous sequential download approach.

Projects

Fast Web Browser | *Rust*

- Implemented a **high-performance** HTML pre-scanner using x86_64 SIMD intrinsics (AVX2) to minimize parsing latency and maximize instruction-level parallelism.
- Designed a software rasterizer writing directly to a pixel framebuffer, including a reusable Canvas abstraction consolidating duplicated rendering logic across 4+ render paths (DOM, diagnostics, debug overlay, chrome).
- Implemented a raw TCP/TLS networking layer with HTTP/1.0 handled from first principles, eliminating reliance on prebuilt HTTP client libraries.

RISC-V Kernel | *C, RISC-V Assembly, QEMU*

- Built a RISC-V kernel from scratch in C, implementing context switching, virtual memory, user mode, a shell, a disk driver, and a file system to produce a fully functional bootable OS.
- Identified and resolved kernel faults by leveraging QEMU's GDB stub and raw memory inspection delivering a stable system.

Technical Skills

Languages: Rust, C/C++, TypeScript, Haskell, Python, Java, Dart, SQL (PostgreSQL, OracleSQL, SQLite), JavaScript
Frameworks: Flutter, React, Node.js, Tokio, Flask, FastAPI, Deno.js
Developer Tools: Git, Docker, perf, eBPF, Cursor, Claude Code, PostgreSQL, Linux (WSL), Xcode, Android Studio
Technologies: FFI, WASI/WASM, P2P Protocols, REST APIs, IMAP, SMTP

Extracurricular Activities

uOttaHack (Student-Led Hackathon)

Community Director

- Helped organize Ottawa's largest annual hackathon (uOttaHack), a Major League Hacking event run in partnership with the University of Ottawa, facilitating workshops, networking, and community-building for 900+ student participants.

Synchronize (Ottawa's Socratica Node)

Host and Organizer

- Host and facilitate weekly co-working sessions for Ottawa's Socratica node, one of 40+ global nodes, bringing together makers, engineers, artists, and designers to work on passion projects and share progress through open demo.